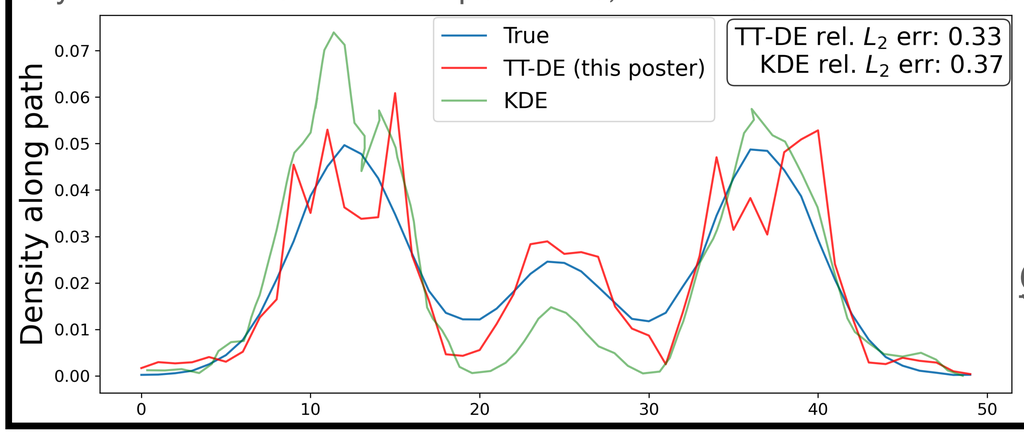
Tensor density estimation in cryo-EM

Matias Andia, Marc Aurèle Gilles

PRINCETON UNIVERSITY Department of Mathematics, Program for Applied and Computational Mathematics

Abstract: In the last decade, low-rank tensor methods have received a lot of attention for density estimation due to their ability to overcome the curse of dimensionality in high dimensions. In this poster, we investigate how the "tensor-train density estimator" (TT-DE), put forth in [2], can be applied to recover the conformational density in cryo-EM datasets. TT-DE significantly outperforms kernel density estimation (KDE) on synthetic datasets and slightly outperforms it on cryo-EM datasets in initial experiments, with much room for further improvements.



github.com/matias-ma/tt-de

The problem

Overview: Density estimation problems are stated as we are given $\left\{z^{(i)}\right\}_{i=1}^{N}$, where $z^{(i)} \in \mathbb{R}^d$ and $z^{(i)} \overset{\text{iid}}{\sim} p$, and our goal is to find $q \approx p$ (according to some measure).

Standard approach: The most straightforward approach, and one that is currently used in many cryo-EM pipelines, is kernel density estimation (KDE), which essentially sums kernels (Gaussians) to fit the density. I.e.:

$q(\boldsymbol{x}) = \frac{1}{N h^d} \sum_{i=1}^{N} K\left(\frac{\boldsymbol{x} - z^{(i)}}{h}\right)$, where K is our (Gaussian) kernel and h is the bandwidth

While KDE works well in low dimensions, its error degrades exponentially with the dimensional d of the space, a phenomenon known as "the curse of dimensionality".

Naive basis expansion: Another idea is to look for an expansion our function w.r.t. a given basis:

$q(\boldsymbol{x}) = \sum_{i_1=1}^{m} \cdots \sum_{i_d=1}^{m} \alpha_{i_1,\ldots,i_d} \phi_{i_1}(x_1) \cdots \phi_{i_d}(x_d)$ where ϕ_j are our univariate basis functions.

The coefficients form a tensor with m^d elements and also suffers from a curse of dimensionality.

Tensor train density estimation (TT-DE)

Background: TT-DE relies on the tensor train (TT) decomposition put forth in [1] to compress the coef. tensor:

$\alpha = G^{(1)} \circ G^{(2)} \cdots \circ G^{(d)}$, where

$\alpha \in \mathbb{R}^{m \times \cdots \times m}$, $G^{(i)} \in \mathbb{R}^{r_{i-1} \times m \times r_i}$, and $\circ$ denotes a contraction. These G's are referred to as the "cores" of α.

Overview: The TT-DE algorithm, proposed in [2], fits the coefficients of the basis expansion estimation described above using an iterative approach while making use of this TT form to overcome the exponential running time.

Cost reduction: Since Φ is separable, it clearly has TT-rank 1. Thus we can calculate $\langle \alpha, \Phi(\boldsymbol{x}) \rangle^2$ in $\mathcal{O}\left(dmr^2\right)$ time and our tensor will have $\mathcal{O}\left(dmr^2\right)$ parameters, thereby making this method's computational and memory costs grow linearly with dimension.

Notation

p is the underlying density

$\hat{p}$ is the empirical density

q is the estimated density

$\alpha \in \mathbb{R}^{m \times \cdots \times m}$ is the coefficient tensor for our basis expansion

$r_0 = r_d = 1$, and $r_1 = \cdots = r_{d-1} = r$ is the "TT-rank" of our tensor/function

m is the number of basis modes

$\Phi(\boldsymbol{x}) = \left\{\phi_k(\boldsymbol{x})\right\}_{k=1}^{m}$ where

$\phi_i(\boldsymbol{x}) = \phi_{i_1,\ldots,i_d}(\boldsymbol{x}) = \phi_{i_1}(x_1) \cdots \phi_{i_d}(x_d)$

and $\phi_{i_k}(x_k)$ is a univariate basis function (we use B-splines)

$\boldsymbol{\phi}(x_i) = \left(\phi_1(x_i), \phi_2(x_i), \ldots, \phi_m(x_i)\right)$

$q(\boldsymbol{x}) = \langle \alpha, \Phi(\boldsymbol{x}) \rangle^2$ where we square it to ensure non negativity

Algorithm

Loss function: TT-DE minimizes the negative log likelihood (NLL):

$\mathscr{L} = -\frac{1}{N} \sum_{i=1}^{N} \log\left(q(z^{(i)})\right)$*

Initialization: It is initialized with a rank-1 density estimator:

$q(\boldsymbol{x}) = \prod_{i=1}^{d} q_i(x_i)$ where

$q_i(x_i) \approx \left\langle \alpha_i, \boldsymbol{\phi}(x_i) \right\rangle$, $\alpha_i \in \mathbb{R}^m$

α_i comes from solving a (1-D) equation $A\alpha_i = b$, where:

$[A]_{j,k} = \langle \phi_j, \phi_k \rangle$ and

$b_k = \langle \hat{p}, \phi_k \rangle$ for $j, k = 1, \ldots, m$

It then solves for q_i, refines it via expectation maximization, and pads it with zeros to get a rank TT-rank r tensor.

Riemannian optimization: TT-DE views the space of tensors with a fixed TT-rank as a Riemannian manifold and runs gradient descent on it: each step moves along the tangent space direction of the gradient, then retracts the point back onto the fixed-rank manifold.

Improvements

Error deconvolution: In cryo-EM, instead of viewing points drawn from the raw density, $z^{(i)} \overset{\text{iid}}{\sim} p$, we observe points drawn from the noise-convolved density, $z^{*(i)} := z^{(i)} + \epsilon^{(i)} \overset{\text{iid}}{\sim} p * U$, where U is the PDF of our noise. So, in order to get a density estimate for f, we will need to change the loss function to be:

$\mathscr{L} = -\frac{1}{N} \sum_{i=1}^{N} \log\left[(U * q)\left(z^{*(i)}\right)\right]$. After this point, the algorithm proceeds the same as outlined earlier. In our initial implementation, the convolution is done via Monte Carlo integration, i.e.:

$(U * q)(\boldsymbol{x}) = \mathbb{E}_{\varepsilon \sim U} q(\boldsymbol{x} - \varepsilon) \approx \frac{1}{M} \sum_{k=1}^{M} q\left(\boldsymbol{x} - \varepsilon^{(k)}\right)$, where $\varepsilon^{(k)} \overset{\text{iid}}{\sim} U$

In future work, we investigate more efficient ways to implement the convolution, e.g if noise kernel has low TT rank and then using a TT-convolution algorithm, such as the one in [3].

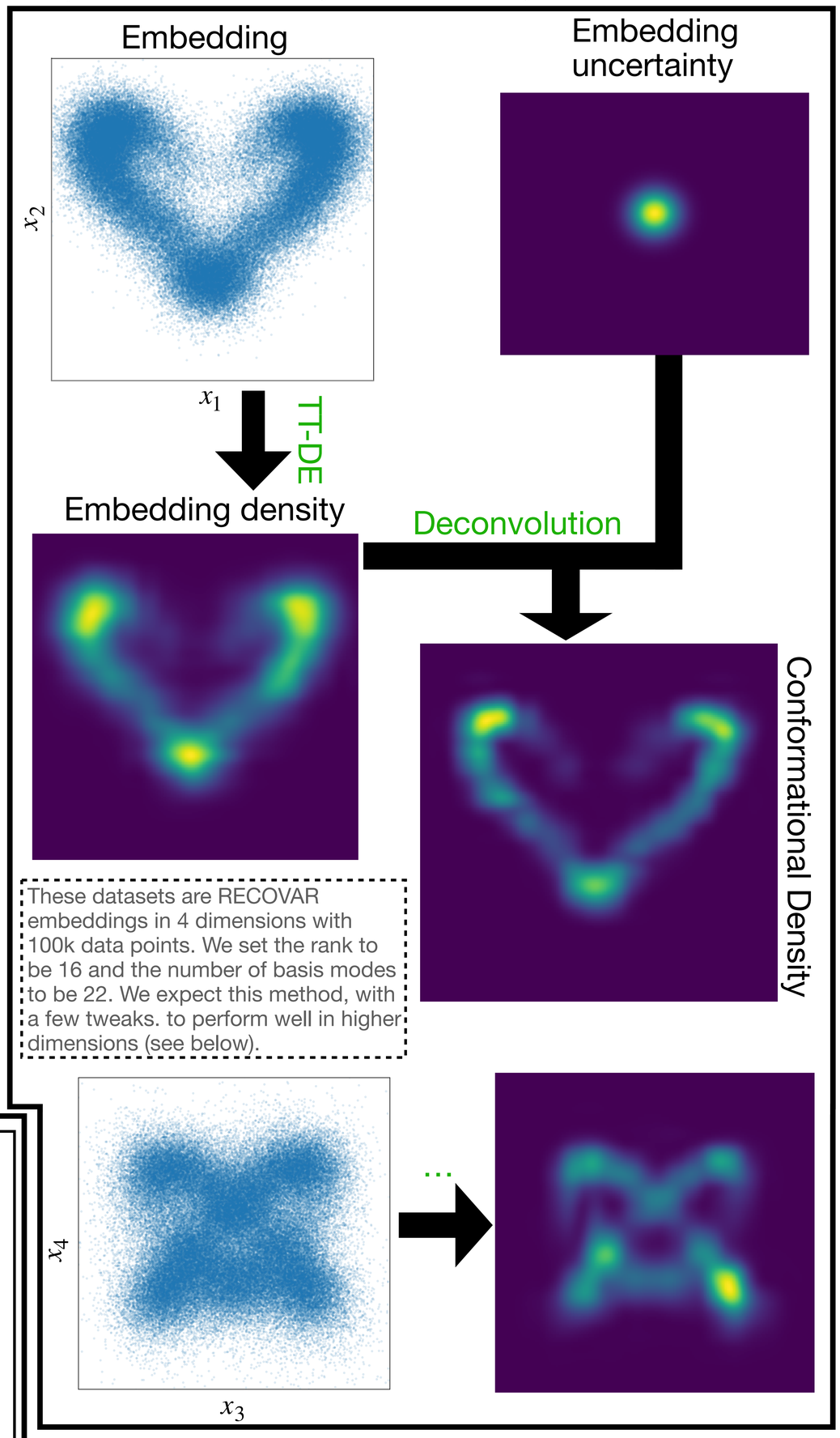
Future additions

Initialization: In [2], the authors emphasize the importance of having a good initialization step for both the convergence speed and the accuracy of TT-DE. Because of this, one promising direction for future research is to implement a non-iterative density estimator built on the TT decomposition (see, e.g., [4]) and employ it in our initialization step.

Regularization: To promote a smooth distribution, in the future we would like to implement the regularization:

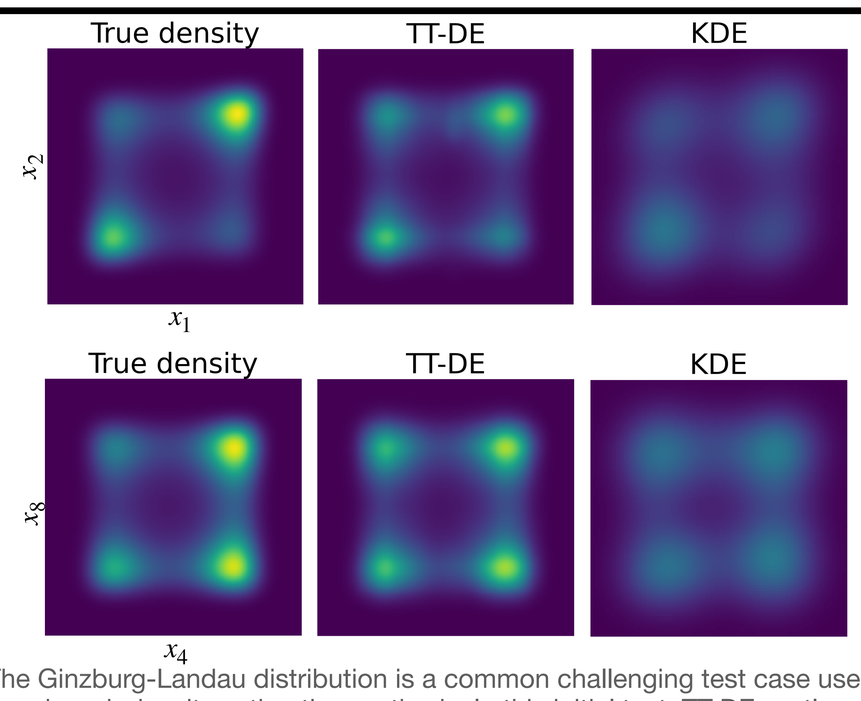
$\mathscr{L}_{\text{reg}} = \mathscr{L} + \lambda \|\nabla q\|_{L_2}^2$ for some regularization parameter $\lambda \in \mathbb{R}^+$

Experiment 1: synthetic cryo-EM data



These datasets are RECOVAR embeddings in 4 dimensions with 100k data points. We set the rank to be 16 and the number of basis modes to be 22. We expect this method, with a few tweaks. to perform well in higher dimensions (see below).

Experiment 2: 10-dimensional Ginzburg-Landau



The Ginzburg-Landau distribution is a common challenging test case used to benchmark density estimation methods. In this initial test, TT-DE vastly outperforms KDE in 10 dimensions, making it a promising candidate to recover high dimensional, complex free-energy landscapes in cryo-EM.

(*) This is different from what they report in their paper (they discuss using the L_2 -error), but it is what they implement in their code.

[1] Oseledets, Ivan V. "Tensor-train decomposition." *SIAM Journal on Scientific Computing* 33.5 (2011).

[2] Novikov, Georgii S., Maxim E. Panov, and Ivan V. Oseledets. "Tensor-train density estimation." *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence (UAI), Proceedings of Machine Learning Research* 161 (2021): 1321–1331.

[3] Rakhuba, M. V. and I. V. Oseledets. "Fast Multidimensional Convolution in Low-Rank Tensor Formats via Cross Approximation." *SIAM Journal on Scientific Computing* 37.2 (2015).

[4] Peng, Yifan, et al. "Tensor Density Estimator by Convolution-Deconvolution." *arXiv* arXiv:2412.18964 (2024).

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